

Smart Traffic Lights Using Local AI: A Decentralized Edge-Computing Approach for Adaptive Traffic and Emergency Vehicle Priority

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Abstract—Traditional traffic-light systems generally operate using predefined timers and have limited ability to respond to rapidly changing traffic conditions. Artificial intelligence (AI) and computer vision provide an opportunity to make traffic signals adaptive by using real-time information obtained from road-side cameras. Recent research has demonstrated the use of object detection, adaptive signal control, and edge computing for intelligent transportation systems.

This paper presents a decentralized smart traffic-light architecture based on local artificial intelligence and edge computing. In the proposed system, street cameras provide traffic information to smart-pole computing nodes located close to the traffic intersection. Local processors perform computer-vision inference and generate traffic-signal decisions with reduced dependence on cloud communication. Selected metadata can simultaneously be transmitted to a city cloud server for long-term storage and traffic analysis. The proposed architecture also incorporates emergency-vehicle detection to support priority control for ambulances and other authorized emergency vehicles. The approach provides a foundation for low-latency and scalable intelligent traffic management.

Index Terms—Smart traffic lights, artificial intelligence, edge computing, computer vision, intelligent transportation systems, adaptive traffic control, emergency vehicle detection, ambulance priority.

I. INTRODUCTION

Urban traffic congestion is an important challenge for modern transportation systems. Conventional traffic lights commonly use fixed-time schedules in which each signal remains active for a predefined duration. Such systems are simple to implement but do not directly respond to sudden variations in traffic demand.

For example, a road may continue to receive a green signal even when only a small number of vehicles are present, while another approach may contain a long queue. This can lead to inefficient utilization of available road capacity.

Intelligent Transportation Systems (ITS) have introduced data-driven methods for improving traffic management. Camera-based computer vision can be used to detect and classify vehicles and estimate traffic conditions in real time. Recent IEEE studies have demonstrated YOLO-based approaches for vehicle detection, classification, counting, and traffic monitoring [1], [2].

AI-based traffic signal control has also been investigated as an alternative to fixed-time control. Object-detection-based

signal logic can estimate vehicle density and dynamically determine green time for different directions [3]. More recent work has combined deep learning and edge computing to perform real-time traffic signal control close to the intersection [4].

A major architectural question is where AI processing should take place. Cloud-based processing provides access to powerful computational resources but requires camera information to be transmitted to a remote server. Edge computing moves computation closer to the data source and can reduce communication requirements for latency-sensitive applications.

Emergency vehicle priority is another important requirement. Recent research has explored AI-based detection of ambulances, fire trucks, and police vehicles for adaptive signal control [5], [6]. However, further work is required to integrate real-time traffic optimization, local edge processing, and emergency priority into a unified architecture.

This paper therefore proposes a decentralized smart traffic-light system in which cameras send traffic information to local smart-pole computers. The local AI system estimates traffic conditions, detects emergency vehicles, and communicates safe signal-control decisions directly to traffic lights.

II. RELATED WORK

A. Traditional Traffic Signal Control

Traditional traffic signals generally operate according to predefined signal cycles. Although fixed-time control is relatively simple, it does not adapt directly to changing traffic conditions.

A fixed signal cycle can be represented as

$$T_{\text{cycle}} = T_{\text{green}} + T_{\text{yellow}} + T_{\text{red}}. \quad (1)$$

The values are normally configured according to expected traffic patterns rather than continuously estimated from current traffic.

B. Computer Vision for Traffic Monitoring

Computer vision provides a non-contact method for estimating traffic conditions. Modern object-detection algorithms can identify vehicles from video streams.

YOLO-based approaches have been applied to real-time traffic monitoring and vehicle classification [1], [2]. Vehicle counts can subsequently be used as inputs to an adaptive signal controller.

C. Adaptive Traffic Signal Control

Object-detection-based traffic signal control has been demonstrated using real-world traffic data. One approach uses YOLO-based vehicle detection to estimate directional traffic load and dynamically calculate green time [3].

Reinforcement-learning approaches have also been proposed for intelligent traffic-light control, where the controller learns a policy from traffic states [7].

Recent work has combined deep learning and edge computing for real-time adaptive traffic signal control. An IEEE study published in 2026 implemented traffic-density estimation on an edge platform and dynamically allocated green-light durations [4].

D. Edge Computing

Edge computing moves computation closer to the location where data is generated. This is particularly useful for applications that require rapid response.

For traffic systems, local processing can reduce the need to continuously transmit raw camera streams to a centralized server. The local node can instead send compact information such as vehicle counts, traffic density, and signal states.

E. Emergency Vehicle Detection

Emergency vehicle detection has become an important research area within intelligent traffic management. Recent work has investigated YOLO-based identification of ambulances, fire engines, and police vehicles in Indian traffic environments [6].

Another recent IEEE study proposed an edge-AI traffic signal system with emergency vehicle detection and priority scheduling [8]. These studies demonstrate the feasibility of integrating emergency vehicle detection with adaptive signal control.

III. PROBLEM STATEMENT

Fixed-time traffic signals cannot efficiently adapt to sudden changes in traffic density. Cloud-based AI systems can provide advanced processing capabilities but may introduce communication overhead and require continuous network connectivity.

Furthermore, emergency situations require rapid coordination between traffic signals. An ambulance approaching an intersection may experience unnecessary delay if the controller is unaware of its presence.

Therefore, this research proposes a decentralized system that:

- 1) detects traffic conditions in real time;
- 2) performs AI inference near the intersection;
- 3) dynamically adjusts signal timing;
- 4) reduces dependence on continuous cloud communication;

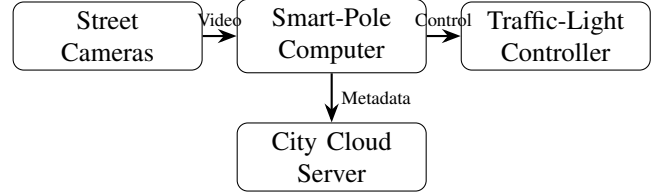


Fig. 1. Proposed decentralized local-AI traffic-light architecture.

- 5) minimizes unnecessary video-data transmission; and
- 6) detects emergency vehicles and provides safe signal priority.

IV. RESEARCH GAP

Existing research demonstrates individual components of intelligent traffic management, including vehicle detection, adaptive signal control, edge computing, and emergency vehicle recognition.

However, there remains an opportunity to develop a unified architecture in which these components operate together at the intersection level.

The main research gap addressed by this paper is the integration of:

- local edge-based AI processing;
- real-time vehicle-density estimation;
- adaptive traffic signal control;
- emergency vehicle detection; and
- selective cloud-based metadata storage.

The proposed architecture therefore places the latency-sensitive decision process at the edge while retaining cloud connectivity for non-real-time analytics.

V. PROPOSED SYSTEM

The proposed system follows a decentralized edge-computing model. Street cameras capture traffic information and provide video input to a smart-pole computer located close to the intersection.

The local computer performs AI inference and determines the traffic state. Based on the estimated traffic density and queue length, the controller dynamically determines the appropriate signal timing.

At the same time, selected metadata can be transmitted to a city cloud server for historical storage and traffic analysis.

The primary control flow is:

Street Cameras → Smart-Pole Computer → Traffic Lights

The secondary data flow is:

Smart-Pole Computer → City Cloud Server

VI. SYSTEM ARCHITECTURE

Fig. 1 shows the proposed architecture.

A. Street Cameras

Street cameras capture the approaching traffic lanes. The captured frames are supplied to the local AI-processing node.

B. Smart-Pole Computer

The smart-pole computer performs local computer-vision inference.

The system can estimate:

- number of vehicles;
- traffic density;
- queue length;
- lane occupancy;
- vehicle category; and
- emergency vehicle presence.

C. Traffic-Light Controller

The controller receives decisions from the local AI node and adjusts the signal phases within predefined safety constraints.

D. City Cloud Server

The cloud server is used for non-critical operations such as:

- historical traffic storage;
- traffic-pattern analysis;
- system monitoring;
- performance analysis; and
- long-term urban traffic planning.

VII. TRAFFIC DENSITY ESTIMATION

Let

$$N_i = \text{number of detected vehicles in lane } i \quad (2)$$

and

$$C_i = \text{estimated capacity of lane } i. \quad (3)$$

The normalized lane occupancy can be estimated as

$$O_i = \frac{N_i}{C_i}. \quad (4)$$

A traffic-demand score can then be calculated as

$$S_i = w_1 N_i + w_2 O_i + w_3 Q_i \quad (5)$$

where Q_i is the queue length and w_1 , w_2 , and w_3 are configurable weights.

The controller can allocate additional green time to approaches with higher demand while maintaining minimum and maximum signal timing constraints.

VIII. EMERGENCY VEHICLE DETECTION AND PRIORITY

Emergency vehicle detection is an important component of the proposed system. YOLO-based approaches have already been explored for identifying ambulances, fire engines, and police vehicles [6].

Let

$$E = \begin{cases} 1, & \text{if an emergency vehicle is detected,} \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

Algorithm 1 Local AI Traffic Signal Control

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1: Capture camera frame
2: Detect and classify vehicles
3: Estimate vehicle density and queue length
4: Check for emergency vehicle
5: if emergency vehicle detected then
6:   Activate safe emergency-priority sequence
7: else
8:   Calculate traffic-demand score
9:   Select highest-demand eligible approach
10:  Adjust green time within safety limits
11: end if
12: Send instruction to traffic-light controller
13: Store selected traffic metadata
14: Forward selected metadata to cloud server

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The controller can use this information to activate an emergency priority procedure.

$$G = \begin{cases} G_{\text{emergency}}, & E = 1, \\ G_{\text{adaptive}}, & E = 0. \end{cases} \quad (7)$$

The emergency procedure must operate within predefined traffic safety constraints. A conflicting signal should not simply be switched to green. Instead, the controller should safely terminate or shorten eligible phases before providing priority.

IX. DECISION ALGORITHM

Algorithm 1 presents the basic operation.

X. CLOUD-BASED AND EDGE-BASED PROCESSING

The latency of a cloud-based architecture can be represented as

$$T_{\text{cloud}} = T_{\text{capture}} + T_{\text{upload}} + T_{\text{inference}} + T_{\text{download}} + T_{\text{control}}. \quad (8)$$

For local processing, the critical control path becomes

$$T_{\text{edge}} = T_{\text{capture}} + T_{\text{local inference}} + T_{\text{control}}. \quad (9)$$

The difference can therefore be expressed as

$$\Delta T = T_{\text{cloud}} - T_{\text{edge}}. \quad (10)$$

The proposed architecture does not claim that every edge system will always outperform every cloud system. Instead, removing network communication from the critical control path can eliminate one source of communication delay.

XI. ARCHITECTURE COMPARISON

Table I compares three conceptual approaches.

The table represents an architectural comparison rather than a controlled experimental benchmark.

XII. ADVANTAGES

The proposed architecture provides several potential advantages.

TABLE I
CONCEPTUAL COMPARISON OF TRAFFIC-CONTROL ARCHITECTURES

Parameter	Fixed	Cloud AI	Local AI
Decision	Timer	Remote AI	Edge AI
Internet dependency	Low	High	Low
Real-time adaptation	No	Yes	Yes
Local processing	No	No	Yes
Cloud analytics	No	Yes	Optional
Emergency detection	No	Possible	Proposed
Raw video transmission	No	High	Reduced

A. Low Communication Dependency

Real-time control can continue even if cloud connectivity is temporarily unavailable.

B. Reduced Data Transmission

Instead of continuously transmitting complete video streams, selected metadata can be transmitted to the cloud.

C. Local Decision Making

The smart-pole computer can perform inference close to the intersection.

D. Adaptive Signal Timing

Signal duration can be adjusted according to observed traffic conditions.

E. Emergency Priority

The same computer-vision infrastructure can detect emergency vehicles and initiate an appropriate priority sequence.

XIII. LIMITATIONS

The system has several limitations.

Computer-vision accuracy may decrease under poor illumination, weather conditions, camera obstruction, and vehicle occlusion. Recent traffic-detection research has specifically reported degradation under low-light and high-occlusion conditions [9].

Local AI hardware also requires computational resources and maintenance.

Emergency-vehicle misclassification could cause inappropriate priority events. Therefore, emergency detection should use suitable confidence thresholds and safety verification.

Finally, AI-generated decisions should not directly bypass deterministic traffic-signal safety logic. The AI system should provide recommendations or commands within a validated controller framework.

XIV. EVALUATION METRICS

The proposed system can be evaluated using the following metrics:

- average vehicle waiting time;
- average queue length;
- intersection throughput;
- AI inference latency;
- emergency detection accuracy;

- emergency response time;
- network bandwidth consumption; and
- number of unsafe signal transitions.

Average waiting time can be calculated as

$$W_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N W_i \quad (11)$$

where W_i represents the waiting time of vehicle i and N is the total number of observed vehicles.

XV. FUTURE WORK

Future work can investigate lightweight object-detection models optimized for edge devices.

Multiple intersections can also be connected to create a distributed traffic-control network.

The system can additionally incorporate:

- vehicle trajectory prediction;
- weather information;
- pedestrian detection;
- vehicle-to-infrastructure communication;
- multi-intersection coordination; and
- federated or distributed learning.

Recent research has investigated end-edge-cloud collaboration and multi-agent learning for regional traffic signal control [10].

XVI. CONCLUSION

This paper presented a decentralized smart traffic-light architecture using local artificial intelligence and edge computing. Unlike fixed-time traffic signals, the proposed system uses camera-based traffic information to dynamically respond to changing road conditions.

The architecture places latency-sensitive computer-vision processing on smart-pole computers located near intersections. Selected metadata can be transmitted to a city cloud server for historical analysis without making cloud communication mandatory for every real-time signal decision.

The proposed system also integrates emergency vehicle detection with adaptive traffic control. Existing research demonstrates the feasibility of YOLO-based vehicle detection, edge-AI traffic control, and emergency vehicle identification [4], [6], [8].

Future experimental evaluation should measure waiting time, queue length, throughput, inference latency, network usage, and emergency response time using real or simulated traffic scenarios.

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